

VARUN DOBHAL

B.Tech Student (Computer Science — Specialization in Cybersecurity)

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Summary

Backend-focused developer with experience in building scalable APIs, secure authentication systems, blockchain applications, and quantum-aware cryptographic systems. Developing real-world platforms with a strong focus on performance optimization, secure, and client requirements. Indian patent holder with an IEEE publication in blockchain security and ongoing research in post-quantum cryptography.

Education

Graphic Era Deemed University

2023 – 2027

B.Tech CSE (Cyber Security)

CGPA: 7.9

Skills

- Languages: JavaScript, Python, Rust, C, Solidity
- Frontend: React.js, Tailwind CSS
- Backend: Node.js, Express, Django, Rocket
- Databases: PostgreSQL, MongoDB, Redis, SQLite
- Tools: Docker, Git, Postman
- Concepts: DSA, OS, DBMS, CN

Work Experience

Freelance Backend Developer

Aug 2025 – Jan 2026

- Built REST APIs handling 1000+ requests/min with Redis caching, reducing latency by 25%
- Implemented JWT authentication and RBAC, securing 500+ user sessions
- Reduced response time by 30% via database optimization

Freelance Full Stack Developer

Oct 2025 – Dec 2025

- Collaborated within a team of 2–3 developers to build full-stack applications using React and Node.js
- Owned the secure authentication module, handling backend development and frontend
- Applied rate limiting, improving system efficiency by 20%

Projects

Independent ERP System for Graphic Era University

Jan 2026 – Present

- Engineering a fully in-house ERP platform and production-grade server infrastructure to support 40K–45K students, reducing reliance on third-party solutions by 100%
- Building scalable student, attendance, and administrative modules capable of processing 5,000+ concurrent requests with optimized database indexing and query performance enhancements.

CTF Platform (National-Level Hackathon)

Feb 2026 – Apr 2026

- Developed a full-stack CTF platform supporting 100+ teams with a real-time WebSocket-based leaderboard, achieving sub-500ms latency for score updates
- Implemented automated flag validation, challenge administration, and proctoring mechanisms to detect and prevent unfair practices during competitions

QR-Based Event Entry Verification System

Jan 2026 – Mar 2026

- Built a high-traffic verification platform for 4,500+ attendees, ensuring zero duplicate entries through a PostgreSQL-backed audit system
- Automated Excel-to-database ingestion and QR generation, reducing the event check-in time per attendee by approximately 60% compared to manual logs

- Architecting a quantum-aware cryptographic framework utilizing adaptive computational hardness to mitigate Shor’s algorithm-based threats
- Developing adaptive cost-escalation mechanisms that increase attack complexity by a factor of 10x against brute-force and quantum-accelerated attempts
- Conducting simulation and benchmarking studies to evaluate security-performance trade-offs in post-quantum environments
- **Authors:** V. Dobhal, M. Wazid

- Constructed JWT-based authentication system supporting 500+ concurrent users
- Initiated rate limiting and API security mechanisms

Publications

- V. Dobhal, S. Mittal, M. Wazid, S. Saha, A. K. Das, S. Pal, J. J. P. C. Rodrigues, “Blockchain-Based Secure Product Authenticity Verification for Industrial Networks,” IEEE ICBC, Brisbane, Australia, June 2026

Patents

- Indian Patent (Published, 2026) — Secure Data Erasure System (202611016955)
- Indian Patent (Published, 2024) — Product Authenticity System (202411082477)

Certifications

• Digital Forensics Essentials (DFE) — EC-Council (Coursera)	Certificate
• Blockchain and its Applications — NPTEL (IIT Kharagpur)	Certificate